2. Colour blindness

In human beings a dominant X – linked gene is necessary for the formation of colour sensitive cells, the cones. The recessive form of this gene is incapable of producing colour sensitive cone cells. Homozygous recessive females (X^cX^c) and hemizygous recessive males (X^cY) are unable to distinguish red and green colour. The inheritance of colour blindness can be studied in the following two types of marriages.

(i) **Marriage between colour blind man and normal visioned woman**

A marriage between a colour blind man and a normal visioned woman will produce normal visioned male and female individuals in F_1 generation but the females are **carriers**. The marriage between a F_1 normal visioned carrier woman and a normal visioned male will produce one normal visioned female, one carrier female, one normal visioned male and one colour blind male in F2 generation. Colour blind trait is inherited from the male parent to his grandson through carrier daughter, which is an example of criss-cross pattern of inheritance **(Fig. 4.3).**

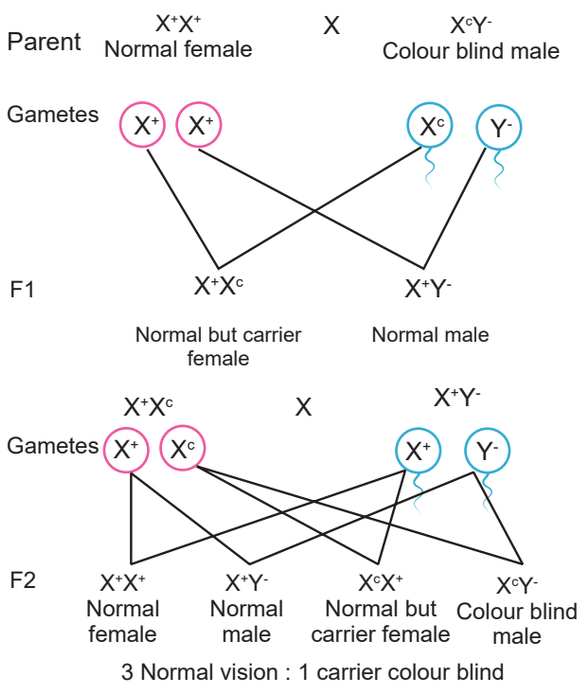


Fig. 4.3 Marriage between colour blind man and normal visioned woman

ii) Marriage between normal visioned man and colour blind woman

If a colour blind woman (X^cX^c) marries a normal visioned male (X^+Y), all F_1 sons will be colourblind and daughters will be normal visioned but are carriers.

Marriage between F_1 carrier female with a **colour blind** male will produce normal visioned carrier daughter, colourblind daughter, normal visioned son and a colourblind son in the F_2 generation **(Fig. 4.4).**

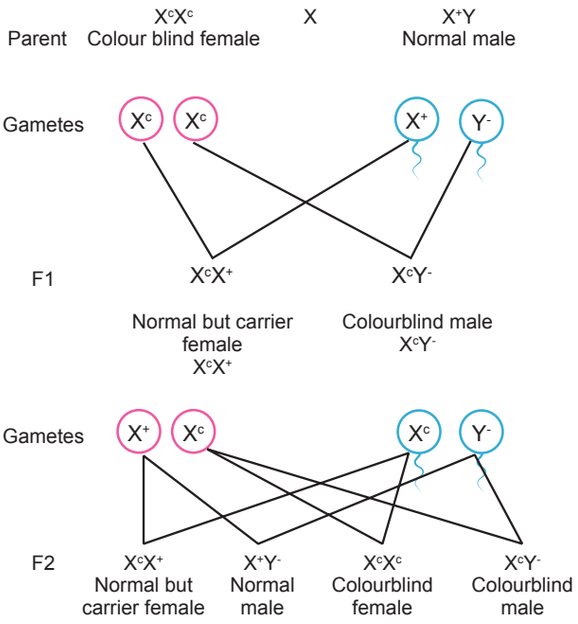


Fig. 4.4 Marriage between normal visioned man and colour blind woman

4.5.2 Inheritance of Y- linked genes

Genes in the non-homologous region of the Y-chromosome are inherited directly from male to male. In humans, the Y-linked or holandric genes for hypertrichosis (excessive development of hairs on pinna of the ear) are transmitted directly from father to son, because males inherit the Y chromosome from the father. Female inherits only X chromosome from the father and are not affected.

4.6 Karyotyping

Karyotyping is a technique through which a complete set of chromosomes is separated

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